

STAT 5200, Notes 8¹

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Readings. Read Chapter 5 in W. These notes cover Sections 5.1, 5.2.1, 5.2.2, 5.2.4, and 5.2.5. In the next set of notes I'll go back to cover Section 5.2.3, resume with 5.2.6, and present examples. Chapter 5 deals with instrumental variables estimation and two-stage least squares. Section 5.1 introduces the topics in a setting where there is only one endogenous explanatory variable. Then Section 5.2 turns to the case where there are several endogenous explanatory variables. It describes the properties and the use of two-stage least squares estimation. We will compute 2SLS estimates by writing code to effect the required steps, rather than by using a canned program. This is a useful approach, because it forces us to confront the details of the procedure. We'll eventually look at examples.

Estimation with a single endogenous explanatory variable and one instrumental variable. Wooldridge begins Chapter 5 with this case. Note that Chapter 5 addresses only a single equation linear model.

Background. Let's start with the model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + u = \mathbf{x}\beta + u. \quad (1)$$

We assume the first $K - 1$ explanatory variables are exogenous, that

$$E[u] = 0, \quad \text{Cov}(x_j, u) = 0, \quad j = 1, \dots, K - 1. \quad (2)$$

The explanatory variable x_K , however, is assumed to be correlated with u . This results in biased estimation of all the regression coefficients in (1), as we have seen.

An instrumental variable. An observable variable z_1 is said to be an *instrumental variable* candidate if two conditions prevail:

$$\text{Cov}(z_1, u) = 0, \quad (3)$$

and the partial correlation between z_1 and x_K is not zero,

$$x_K = \delta_0 + \delta_1 x_1 + \cdots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + r_K, \quad \text{with } \theta_1 \neq 0. \quad (4)$$

¹This is based on Paul Shaman's lecture notes (Notes 8 from STAT 5200) at the University of Pennsylvania. Last revised: Summer, 2024. Last compiled: November 19, 2025. Lecture notes and associated materials are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license.

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For condition (4) we require that the endogenous explanatory variable be projected onto the constant variable, all the exogenous explanatory variables, and the IV. Equation (4) is called a *reduced form equation* for the endogenous explanatory variable x_K . Sometimes estimates of the parameters in a reduced form equation such as (4) are of interest in themselves and are reported. Usually we are more interested in estimating the parameters in equation (1). From (1) and (4) we can write

$$\begin{aligned} y &= (\beta_0 + \beta_K \delta_0) + (\beta_1 + \beta_K \delta_1)x_1 + \cdots + (\beta_{K-1} + \beta_K \delta_{K-1})x_{K-1} + \beta_K \theta_1 z_1 + u + \beta_K r_K \\ &= \alpha_0 + \alpha_1 x_1 + \cdots + \alpha_{K-1} x_{K-1} + \lambda_1 z_1 + v. \end{aligned} \quad (5)$$

The requirements for an instrumental variable are given by (3) and (4). We could strengthen (3) to require $E[u|z_1] = 0$. We also talk about the *exclusion restriction* for the instrumental variable. This states that z_1 does not directly impact y – it affects y only via its correlation with x_K . I'll discuss this more fully later in these notes.

Identification. Recall that a model is said to be *identified* if its parameters can be expressed in terms of population moments of observable variables. Let's demonstrate that the model (1) is identified if the assumptions (2) and (3) specified for the IV z_1 hold. Write

$$\mathbf{x} = (1, x_1, \dots, x_{K-1}, x_K), \quad \mathbf{z} = (1, x_1, \dots, x_{K-1}, z_1) \quad (6)$$

Multiply $y = \mathbf{x}\beta + u$ on the left by $\mathbf{z}'$ and take the expected value

$$E[\mathbf{z}'\mathbf{x}\beta + \mathbf{z}'u] = E[\mathbf{z}'\mathbf{x}]\beta + 0 = E[\mathbf{z}'y].$$

We assume that $E[\mathbf{z}'\mathbf{z}]$ has full rank. This precludes redundancies in the $\mathbf{z}$ vector and is thus a reasonable assumption. The model is identified if the $(K+1) \times (K+1)$ matrix $E[\mathbf{z}'\mathbf{x}]$ has rank $K+1$. This rank condition holds if and only if $\theta_1 \neq 0$ in (4), and this is proved via Problem 5.12 in W. Under the latter rank condition we have

$$\beta = [E[\mathbf{z}'\mathbf{x}]]^{-1} E[\mathbf{z}'y]. \quad (7)$$

The instrumental variable estimator of β . Using the data, define the $N \times (K+1)$ matrices

$$\mathbf{Z} = \begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \vdots \\ \mathbf{z}_N \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{pmatrix} \quad (8)$$

and the $N \times 1$ vector

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}. \quad (9)$$

Then the IV estimator of β is (using observed values of $\mathbf{x}$, $\mathbf{z}$, and y)

$$\hat{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{z}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{z}'_i y_i \right) = (\mathbf{Z}'\mathbf{X})^{-1} \mathbf{Z}'\mathbf{y}. \quad (10)$$

The estimator (10) is consistent by the law of large numbers and Lemma 3.4 in Wooldridge.

Thus, the problem of obtaining a consistent estimator for the parameter vector is solved by simply replacing the endogenous explanatory variable x_K by the instrumental variable z_1 to form $\mathbf{z}$ as in (6), and then constructing (10). Essentially, this is a one-step procedure.

There is always the matter of finding instruments for endogenous explanatory variables. Wooldridge discusses several examples on pages 93–96. Note that the assumption shown in (4) can and should be tested (with a t statistic). That is, one should perform the regression in (4) and test the hypothesis $H_0 : \theta_1 = 0$. If this hypothesis is not rejected, then z_1 should not be employed as an instrumental variable for x_K . One should employ care in performing this test – it may be necessary to use a heteroscedasticity-robust t statistic.

Note that an instrumental variable and a proxy variable are different. If there is an endogenous explanatory variable because the structural equation has an omitted variable, we want our instrumental variable to be uncorrelated with the omitted variable. By contrast, a proxy variable should be *correlated* with the omitted variable.

Multiple instruments and two-stage least squares for a single endogenous explanatory variable. For now let's continue to assume there is one endogenous explanatory variable, x_K . And we are considering the model (1). However, now suppose that we have multiple candidate instruments for x_K , the variables $z_1, \dots, z_M$. These satisfy

$$\text{Cov}(z_j, u) = 0, \quad j = 1, \dots, M. \quad (11)$$

In addition, because we have identified these variables as candidate instruments, we expect each of them to have nonzero partial correlation with x_K . More about this below.

When a model has more instruments than there are endogenous explanatory variables, the model is said to be *overidentified*. If the number of instruments exactly matches the number of endogenous explanatory variables, the model is said to be *just-identified*.

Each of the M z_j 's can produce an IV estimator of the β vector, and so can any linear combination of the z_j 's (any linear combination is uncorrelated with u). The *two-stage least squares* (2SLS)

procedure chooses the linear combination of the z_j 's which has maximum correlation with x_K (see the Appendix). Define the vector of exogenous variables,

$$\mathbf{z} = (1, x_1, \dots, x_{K-1}, z_1, \dots, z_M), \quad (12)$$

a $1 \times (K + M)$ vector. The reduced form model for x_K is

$$x_K = \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M + r_K, \quad (13)$$

where the error term r_K has mean 0 and is uncorrelated with the x_j 's and the z_j 's. The maximum correlation linear combination is then

$$x_K^* = \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M. \quad (14)$$

We can't observe x_K^* , of course, and thus it can't be used as an instrument for x_K . So we estimate the parameters in (13) by OLS and use as an instrument the OLS fitted values,

$$\hat{x}_{iK} = \hat{\delta}_0 + \hat{\delta}_1 x_{i1} + \dots + \hat{\delta}_{K-1} x_{iK-1} + \hat{\theta}_1 z_{i1} + \dots + \hat{\theta}_M z_{iM}, \quad i = 1, \dots, N. \quad (15)$$

Define, analogously to $\mathbf{z}$ in (6),

$$\hat{\mathbf{x}} = (1, x_1, \dots, x_{K-1}, \hat{x}_K). \quad (16)$$

Then the *two-stage least squares estimator* of the parameter vector is

$$\hat{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \hat{\mathbf{x}}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \hat{\mathbf{x}}'_i y_i \right) = (\hat{\mathbf{X}}' \mathbf{X})^{-1} \hat{\mathbf{X}}' \mathbf{y}. \quad (17)$$

Compare (10).

The estimator (17) is obtained from a two-stage least squares procedure:

- (i) Regress the endogenous explanatory variable x_K on $1, x_1, \dots, x_{K-1}, z_1, \dots, z_M$ and form the variable $\hat{x}_K$ (giving the fitted values). This is the first-stage regression. It is important to include all of the exogenous explanatory variables $x_1, \dots, x_{K-1}$ in this regression (see Problem 5.11 in W, which I will discuss in class). We can think of $\hat{x}_K$ as the exogenous portion of x_K .
- (ii) Regress y on $1, x_1, \dots, x_{K-1}, \hat{x}_K$. This is the second-stage regression, and it yields the estimator (17).

Let's further explore the form of the estimator (17) and show that it is in fact an OLS estimator (the result we obtain will clarify what this means). The $N \times (K + 1)$ matrix $\mathbf{X}$ is defined at (8). Let's express it in terms of its columns as

$$\mathbf{X} = (\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_{K-1}, \mathbf{X}_K),$$

where $\mathbf{1}$ is an $N \times 1$ vector of 1's. Analogously, using (12), define the $N \times (K + M)$ matrix $\mathbf{Z}$ in terms of its columns by

$$\mathbf{Z} = (\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_{K-1}, \mathbf{Z}_1, \dots, \mathbf{Z}_M),$$

and assume that $\mathbf{Z}$ has column rank $K + M$. Also define the matrix $\hat{\mathbf{X}}$ in terms of its columns and rows,

$$\hat{\mathbf{X}} = (\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_{K-1}, \hat{\mathbf{X}}_K) = \begin{pmatrix} \hat{\mathbf{x}}_1 \\ \vdots \\ \hat{\mathbf{x}}_N \end{pmatrix}.$$

Further, define the parameter vector in (13) as

$$\begin{pmatrix} \delta \\ \theta \end{pmatrix} = (\delta_0, \delta_1, \dots, \delta_{K-1}, \theta_1, \dots, \theta_K)'$$

The OLS estimator of this parameter vector [from the first stage of 2SLS, as described above; see (13) and (15)] is $(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}_K$, and thus

$$\hat{\mathbf{X}}_K = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}_K = \mathbf{P}\mathbf{X}_K, \quad \mathbf{P} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'. \quad (18)$$

The matrix $\mathbf{P}$ is symmetric and idempotent – it is a projection matrix. It is also true that

$$\hat{\mathbf{X}} = \mathbf{P}\mathbf{X}, \quad (19)$$

because

$$\mathbf{P}\mathbf{1} = \mathbf{1}, \quad \mathbf{P}\mathbf{X}_j = \mathbf{X}_j, \quad j = 1, \dots, K - 1$$

(why? I will discuss this in class). Then in (17)

$$(\hat{\mathbf{X}}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{P}'\mathbf{X})^{-1} = (\mathbf{X}'\mathbf{P}\mathbf{X})^{-1} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}.$$

Thus, the estimator (17) can be written as

$$\hat{\beta} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y}. \quad (20)$$

The estimators (10) and (17) are identical when there is only one instrument for x_K . This will be clarified below.

Let's return to the comment in the two lines below (11). For this discussion $\mathbf{z}$ is defined by (12) and $\mathbf{x}$ is defined by (6). That is, we have multiple instruments for one endogenous explanatory variable. For the model to be identified, we don't require each of the candidate instrumental variables to have nonzero partial correlation with x_K . It is only necessary that at least one of the θ_j in (13) be nonzero. An equivalent condition is that the $(K + M) \times (K + 1)$ matrix $E[\mathbf{z}'\mathbf{x}]$ have column rank $K + 1$. (See Problem 5.12 in W.) We test this rank condition by testing the null hypothesis $H_0 : \theta_1 = \dots = \theta_M = 0$ vs. the alternative that at least one of the θ_j 's is nonzero. This is

implemented with a partial F test, and one also needs to determine whether a heteroscedasticity-robust procedure is required. If the null hypothesis is not rejected, the instruments do not satisfy a requirement and the 2SLS procedure should probably not be employed. We also need to be concerned about weak instruments (more about this later).

As noted above, when there is a single endogenous explanatory variable and $M > 1$, the model is said to be overidentified. There are then said to be $M - 1$ overidentifying restrictions.

2SLS generally. In the general situation there are multiple endogenous explanatory variables and multiple instrumental variables.

Assumptions. We consider the model $y = \mathbf{x}\beta + u$. We assume for this discussion that $\mathbf{x}$ includes 1 and is $1 \times K$.

Assumption 2SLS.1. We have observed values of a $1 \times L$ vector $\mathbf{z}$ for which $E[\mathbf{z}'u] = \mathbf{0}$.

The first element of $\mathbf{z}$ is 1, and $\mathbf{z}$ contains the exogenous variables in $\mathbf{x}$ plus instrumental variables from outside the model. The next assumption imposes a rank condition.

Assumption 2SLS.2. $E[\mathbf{z}'\mathbf{z}]$ has rank L ; $E[\mathbf{z}'\mathbf{x}]$ has (column) rank K .

Note that $E[\mathbf{z}'\mathbf{z}]$ is $L \times L$ and $E[\mathbf{z}'\mathbf{x}]$ is $L \times K$. Thus we must have $L \geq K$; that is, we must have at least as many instruments as there are explanatory variables (the exogenous explanatory variables are their own instruments). The first rank condition in this assumption will be satisfied if the instruments have been chosen well (of course, we can assume the exogenous explanatory variables in $\mathbf{x}$ are not collinear). The second part of the assumption is the rank condition required for the model to be identifiable. Note that $E[\mathbf{z}'\mathbf{x}]$ has full column rank if $\mathbf{z}$ is sufficiently linearly related to $\mathbf{x}$.

If all the elements of $\mathbf{x}$ are exogenous, then $\mathbf{z} = \mathbf{x}$ and these two assumptions reduce to the assumptions OLS.1 and OLS.2 on page 2 of Notes 3.

An expression for the 2SLS estimator of the parameter vector. In this general setting the 2SLS estimator of β can be written as in (20). In the matrix $\widehat{\mathbf{X}}$ each of the endogenous explanatory variable columns is replaced by its prediction from the first stage estimation. Let's write (20) in a manner that helps clarify the form of the estimator. The expressions in (18) and (19) are applicable, and we have

$$\widehat{\mathbf{X}}'\widehat{\mathbf{X}} = \mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{X} = \mathbf{X}'\mathbf{P}\mathbf{X} = \mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}, \quad \widehat{\mathbf{X}}'\mathbf{y} = \mathbf{X}'\mathbf{P}'\mathbf{y} = \mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}. \quad (21)$$

Remember that $\mathbf{Z}$ contains the exogenous explanatory variables and the instrumental variables. Therefore,

$$\begin{aligned}\hat{\beta} &= (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y} \\ &= [\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}]^{-1}\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} \\ &= \left[\left(\sum_{i=1}^N \mathbf{x}_i'\mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{x}_i \right) \right]^{-1} \left(\sum_{i=1}^N \mathbf{x}_i'\mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i'y_i \right).\end{aligned}\tag{22}$$

Substituting $\mathbf{x}_i\beta + u_i$ for y_i in (22), we obtain

$$\begin{aligned}\hat{\beta} &= \beta + \left[\left(\sum_{i=1}^N \mathbf{x}_i'\mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{x}_i \right) \right]^{-1} \left(\sum_{i=1}^N \mathbf{x}_i'\mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i'\mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i'u_i \right) \\ &= \beta + (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{u},\end{aligned}\tag{23}$$

where $\mathbf{u} = (u_1, \dots, u_N)'$. Applying the law of large numbers and Lemma 3.4 in Wooldridge, we see that $\hat{\beta}$ consistently estimates β .

Let's revisit Assumption 2SLS.2, the rank condition. We have stated that the model is identified if this condition holds. To see this, follow the steps leading to (22) *in the population*, without performing the matrix inversion for the triple product on the left in (22). This yields

$$E[\mathbf{x}'\mathbf{z}] [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{x}]\beta = E[\mathbf{x}'\mathbf{z}] [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'y],$$

and the triple product on the left is an invertible matrix if and only if $E[\mathbf{z}'\mathbf{x}]$ has column rank K .

Comment 1. Rather than explicitly showing the two stages of the estimation leading to $\hat{\beta}$, (22) expresses the estimator directly in terms of the explanatory variable values $\mathbf{x}_i$, the exogenous variable values $\mathbf{z}_i$, and the response variable values y_i . If we consider the two stages, we see that the first stage of 2SLS regresses each endogenous explanatory variable on the exogenous explanatory variables and the instrumental variables collectively. This yields predictions for the endogenous explanatory variables. These predictions replace the values of the endogenous explanatory variables in the $\mathbf{x}_i$ vectors, yielding $\hat{\mathbf{x}}_i$ vectors. Finally, the estimator (22) is obtained from the second stage regression, that of y vs. $\hat{\mathbf{x}}$.

Comment 2. The conditions (3) and (11) are called exclusion restrictions. If they hold, the instruments are excluded from the original model, because adding the instruments to this model would not capture any additional variation in the response variable. That is, the instruments cannot be explanatory variables in the original equation. The instruments impact the response

variable only indirectly, through their correlation with the endogenous explanatory variables for which they are instruments, and this is made explicit by the first stage equations.

Asymptotic normality of the 2SLS estimator, and the asymptotic variance of the estimator. First note that

$$\sqrt{N}(\hat{\beta} - \beta) = \left[E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) \right]^{-1} E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{z}'_i u_i \right) + o_p(1)$$

Note that by CLT, we have

$$\left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{z}'_i u_i \right) \rightarrow_d N(0, E[u_i^2 \mathbf{z}'_i \mathbf{z}_i]).$$

Therefore, putting altogether we have

$$\sqrt{N}(\hat{\beta} - \beta) \rightarrow_d N(0, V_{asy})$$

where

$$\begin{aligned} V_{asy} &= \left[E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) \right]^{-1} \\ &\quad \times E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E[u_i^2 \mathbf{z}'_i \mathbf{z}_i] E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) \\ &\quad \times \left[E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) \right]^{-1}. \end{aligned}$$

The next assumption is that of homoscedasticity.

Assumption 2SLS.3. $E[u^2 \mathbf{z}' \mathbf{z}] = \sigma^2 E[\mathbf{z}' \mathbf{z}]$, where $E[u^2] = \sigma^2$.

In Notes 3 we made this assumption with $\mathbf{x}$ instead of $\mathbf{z}$. Note that a sufficient condition for this assumption is $E[u^2 | \mathbf{z}] = \sigma^2$.

In the population, the vector of linear projections of $\mathbf{x}$ onto $\mathbf{z}$ is $\mathbf{x}^* = \mathbf{z}\Pi$, where Π is the $L \times K$ matrix equal to $[E[\mathbf{z}' \mathbf{z}]]^{-1} E[\mathbf{z}' \mathbf{x}]$.

Under the Assumptions 2SLS.1–3. $\sqrt{N}(\hat{\beta} - \beta)$ converges in distribution to a normal distribution with mean $\mathbf{0}$ and covariance matrix

$$\sigma^2 \left[E[\mathbf{x}' \mathbf{z}] [E[\mathbf{z}' \mathbf{z}]]^{-1} E[\mathbf{z}' \mathbf{x}] \right]^{-1} = \sigma^2 \left[E[\mathbf{x}^* \mathbf{x}^*] \right]^{-1}. \quad (24)$$

This is because the middle term in V_{asy} becomes

$$\begin{aligned} E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E[u_i^2 \mathbf{z}'_i \mathbf{z}_i] E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) &= \sigma^2 E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E[\mathbf{z}'_i \mathbf{z}_i] E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i) \\ &= \sigma^2 E(\mathbf{x}'_i \mathbf{z}_i) E(\mathbf{z}'_i \mathbf{z}_i)^{-1} E(\mathbf{z}'_i \mathbf{x}_i). \end{aligned}$$

Note that the terms, $E[\mathbf{z}'_i \mathbf{z}_i] E(\mathbf{z}'_i \mathbf{z}_i)^{-1}$, in the middle of the right-hand-side of the first line cancels each other.

To estimate σ^2 , we use the residuals from the 2SLS procedure,

$$\hat{u}_i = y_i - \mathbf{x}_i \hat{\beta}, \quad i = 1, \dots, N,$$

and form

$$\hat{\sigma}^2 = \frac{1}{N-K} \sum_{i=1}^N \hat{u}_i^2. \quad (25)$$

Be careful with this estimation; don't use the residuals from the second stage regression—they are equal to $y_i - \hat{\mathbf{x}}_i \hat{\beta}$. Sometimes the divisor N is used in (25). For asymptotic purposes the distinction doesn't matter, but one has to be concerned about poor approximation in finite samples.

The estimator of the $K \times K$ asymptotic covariance matrix of $\hat{\beta}$ is then given by an estimator of (24) divided by N ,

$$\hat{\sigma}^2 \left(\sum_{i=1}^N \hat{\mathbf{x}}'_i \hat{\mathbf{x}}_i \right)^{-1} = \hat{\sigma}^2 (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1}, \quad (26)$$

or, what is equivalent,

$$\hat{\sigma}^2 \left[\left(\sum_{i=1}^N \mathbf{x}'_i \mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}'_i \mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}'_i \mathbf{x}_i \right) \right]^{-1} = \hat{\sigma}^2 [\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X}]^{-1}. \quad (27)$$

The standard errors of the individual coefficient estimators are then the square roots of the diagonal entries of the matrices (26) and (27). These expressions assume homoscedasticity. Let's consider the asymptotic covariance matrix of $\hat{\beta}$ when there is heteroscedasticity (we don't use Assumption 2SLS.3). We consult (21)–(23) and see that the asymptotic covariance matrix is estimated by

$$\begin{aligned} &(\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \left(\sum_{i=1}^N \hat{u}_i^2 \hat{\mathbf{x}}'_i \hat{\mathbf{x}}_i \right) (\hat{\mathbf{X}}' \hat{\mathbf{X}})^{-1} \\ &= [\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X}]^{-1} \mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \sum_{i=1}^N \hat{u}_i^2 \mathbf{z}'_i \mathbf{z}_i (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X} [\mathbf{X}' \mathbf{Z} (\mathbf{Z}' \mathbf{Z})^{-1} \mathbf{Z}' \mathbf{X}]^{-1}. \end{aligned} \quad (28)$$

One can modify (28) in the manner of HC1–HC4 of the sandwich package in R.

As an alternative to (27), one can use the bootstrap and the jackknife to obtain standard errors of the estimates.

Hypothesis testing and confidence intervals. Hypothesis tests, confidence intervals and confidence regions are constructed in the usual manner, based on t and F distributions (or, chi-squared distribution), with standard errors determined from (27) or (28). Or one can use the Wald statistic for hypothesis testing. And one can use the bootstrap to construct confidence intervals.

Computation with R. One can use (22) to obtain the 2SLS estimates of the parameter vector. Here's an alternative sequence of steps which is reasonably economical.

1. (first-stage regressions) Regress each endogenous explanatory variable x_j vs. the exogenous explanatory variables and the chosen instrumental variables. From each such regression form the predicted (fitted) values of the endogenous explanatory variable, that is, form $\hat{x}_j$. See (15) above.
2. (second-stage regression) Replace each endogenous explanatory variable, x_j , by its predicted values, that is, by $\hat{x}_j$. Regress the response variable vs. the exogenous explanatory variables and all of the $\hat{x}_j$'s formed in step 1. This performs the calculation in (22) [see the first line of (22)]. That is, it correctly calculates the 2SLS estimate of the regression vector β . However, the asymptotic standard errors of the estimates from this second-stage regression are off.
3. (asymptotic standard errors assuming homoscedasticity) The correct standard errors for the 2SLS estimates are obtained from (25) and (26), where the residual values are $\hat{u}_i = y_i - \mathbf{x}_i\hat{\beta}$, $i = 1, \dots, N$.
4. (asymptotic standard errors assuming heteroscedasticity) As noted above, one uses (28), possibly with the modifications in the manner of HC1–HC4 of the sandwich package in R.

Additional discussion 1 (single explanatory variable, single IV). Let's consider a simple setting. Suppose the causal model of interest is

$$y = \beta_0 + \beta_1 s + u, \tag{29}$$

where y is the log of the wage and s is years of schooling. We assume the correlation between s and u is not 0 because ability is a missing variable absorbed in the error term u , and s and ability are correlated. The variable s is endogenous. If we use (29) to perform OLS estimation of β_0 and β_1 , the estimates will be biased (asymptotically biased).

Assume that z is an instrumental variable for this situation. Specifically, z is uncorrelated with u , and it is correlated with s . We can write

$$s = \theta_0 + \theta_1 z + r, \tag{30}$$

with $\theta_1 \neq 0$, and r has mean 0 and is uncorrelated with z . To obtain unbiased estimation of the parameters, we replace s in (29) by its linear projection

$$\hat{s} = L(s|1, z) = \hat{\theta}_0 + \hat{\theta}_1 z$$

and perform OLS estimation.

It is useful to repeat Comment 2 in the earlier part in the present context.

- The instrumental variable z does not affect the outcome y other than through its association with s . It is not part of the causal model, and it does not add anything to knowledge of y beyond that which is already provided by s . That is, $E[y|s, z] = E[y|s]$.
- In the econometric literature the phrase *exclusion restriction* is used to describe the fact that the error term in a causal model is uncorrelated with the chosen instrument. When there is lack of such correlation, there is no benefit to be gained by adding the instrument to the causal model – its addition cannot capture any more of the variation in the response variable (what the first bullet states).

Let's reexamine (7) on page 2 in the present context. According to (7), the IV estimator of the parameters in (29) is, with $\mathbf{z} = (1 \ z)$ and $\mathbf{s} = (1 \ s)$, obtained from the sample analog of

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \left[E \begin{pmatrix} 1 \\ z \end{pmatrix} (1 \ s) \right]^{-1} E \begin{pmatrix} 1 \\ z \end{pmatrix} y = [E[\mathbf{z}'\mathbf{s}]]^{-1} E[\mathbf{z}'y] = \left[[E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{s}] \right]^{-1} \times [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'y]. \quad (31)$$

The factor $[E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'y]$ on the far right of (31) is the 2×1 vector of regression coefficients from the population linear projection of y onto $\mathbf{z} = (1 \ z)$ (the reduced form). And the factor $[E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{s}]$ is the 2×2 matrix of regression coefficients from the population linear projection of $\mathbf{s} = (1, \ s)$ onto $\mathbf{z} = (1 \ z)$ (the first stage regression).

The middle expression in (31) yields

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = [E[\mathbf{z}'\mathbf{s}]]^{-1} E[\mathbf{z}'y] = \begin{pmatrix} 1 & E[s] \\ E[z] & E[sz] \end{pmatrix}^{-1} \begin{pmatrix} E[y] \\ E[yz] \end{pmatrix} = \frac{1}{\text{cov}(s, z)} \begin{pmatrix} E[sz] & -E[s] \\ -E[z] & 1 \end{pmatrix} \begin{pmatrix} E[y] \\ E[yz] \end{pmatrix}. \quad (32)$$

The second component in (32) is

$$\beta_1 = \frac{\text{cov}(y, z)}{\text{cov}(s, z)} = \frac{\text{cov}(y, z)/\text{var}(z)}{\text{cov}(s, z)/\text{var}(z)}. \quad (33)$$

Compare (33) and the expression on the far right of (31). In a more general setting,

$$y = \mathbf{x}\beta + u,$$

some of the variables in $\mathbf{x}$ are exogenous and some are endogenous, and $\mathbf{z}$ is a vector of instrumental variables consisting of the exogenous variables (they are their own instruments) and chosen instruments for the endogenous variables. Then (7) is

$$\beta = [E[\mathbf{z}'\mathbf{x}]]^{-1} E[\mathbf{z}'y] = \left[[E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{x}] \right]^{-1} [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'y]. \quad (34)$$

Additional discussion 2: The Wald estimator. In 1940, there appeared a paper by Abraham Wald deriving an unbiased estimator of the slope coefficient in a simple regression when both the dependent variable and the independent variable are measured with error [Wald, A. (1940). The fitting of straight lines if both variables are subject to error. *The Annals of Mathematical Statistics* **11** 284–300]. Recall that measurement error in independent variables tends to shrink estimates of regression coefficients toward zero. Interestingly, the estimator Wald proposed also arises in the context of a simple regression when the explanatory variable is endogenous and a binary variable is instrumented for it.

We write the simple regression model as

$$y = \beta_0 + \beta_1 x + u, \quad (35)$$

where the single explanatory variable x is endogenous. The instrumental variable z for x is binary, equal to 1 and 0. From (33), the slope in (35) is

$$\beta_1 = \frac{\text{cov}(y, z)}{\text{cov}(x, z)}, \quad (36)$$

and we need to evaluate this ratio when z is binary. Assume $P(z = 1) = p$, $P(z = 0) = 1 - p$. Then $E[z] = p$. The numerator in (36) is

$$\begin{aligned} \text{cov}(y, z) &= E[yz] - E[y]E[z] \\ &= E[yz|z=1]P(z=1) + E[yz|z=0]P(z=0) \\ &\quad - [E[y|z=1]P(z=1) + E[y|z=0]P(z=0)] E[z] \\ &= E[y|z=1]p - [E[y|z=1]p + E[y|z=0](1-p)] p \\ &= [E[y|z=1] - E[y|z=0]] p(1-p). \end{aligned}$$

Similarly,

$$\text{cov}(x, z) = [E[x|z=1] - E[x|z=0]] p(1-p).$$

Then (36) is

$$\beta_1 = \frac{E[y|z=1] - E[y|z=0]}{E[x|z=1] - E[x|z=0]}. \quad (37)$$

Let's calculate the sample analog of (36) and (37). Assume we have N independent sample values (y_i, x_i, z_i) , $i = 1, \dots, N$, and that N_0 of the z_i 's are equal to 0 and N_1 of them are equal to 1 ($N_0 + N_1 = N$). Moreover, $\bar{y}_0$ is the sample mean of the response values for which $z_i = 0$, $\bar{y}_1$ is the sample mean of the response values for which $z_i = 1$, and $\bar{x}_0$ and $\bar{x}_1$ are defined similarly. Using (36), we find that the estimator of the slope in (35) is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (y_i - \bar{y})(z_i - \bar{z})}{\sum_{i=1}^N (x_i - \bar{x})(z_i - \bar{z})} = \frac{\sum_{i=1}^N (y_i - \bar{y})z_i}{\sum_{i=1}^N (x_i - \bar{x})z_i} = \frac{N_1\bar{y}_1 - \frac{N_1(N_0\bar{y}_0 + N_1\bar{y}_1)}{N_0 + N_1}}{N_1\bar{x}_1 - \frac{N_1(N_0\bar{x}_0 + N_1\bar{x}_1)}{N_0 + N_1}} = \frac{\bar{y}_1 - \bar{y}_0}{\bar{x}_1 - \bar{x}_0}. \quad (38)$$

This is the Wald estimator. See Problem 5.13 in W.

One final comment: There is an alternative derivation of (37). Assume the stronger condition

$$E[u|z] = 0.$$

Then, from (35), we can write

$$E[y|z] = \beta_0 + \beta_1 E[x|z]. \quad (39)$$

This is the same as

$$E[y|z=1]P(z=1) + E[y|z=0]P(z=0) = \beta_0 + \beta_1 [E[x|z=1]P(z=1) + E[x|z=0]P(z=0)].$$

This is equivalent to

$$\begin{aligned} [E[y|z=1] - E[y|z=0]]p &= \beta_1 [E[x|z=1] - E[x|z=0]]p \\ &\quad - [E[y|z=0] - \beta_0 - \beta_1 E[x|z=0]]. \end{aligned} \quad (40)$$

We see that the second line of (40) is zero, by conditioning on $z = 0$ in (39). Thus, solving for β_1 in (40), we obtain (37).

Appendix (optional). ³ Suppose y is a random variable and $\mathbf{x}$ is a $1 \times K$ random vector. The maximum correlation between y and the linear combination $\mathbf{x}\gamma$, where γ is $K \times 1$, is called the *multiple correlation coefficient* between y and $\mathbf{x}$.

Theorem. Among all linear combinations $\mathbf{x}\gamma$, the linear projection $L(y|\mathbf{x}) = \mathbf{x}\beta$, where $\beta = [E[\mathbf{x}'\mathbf{x}]]^{-1} E[\mathbf{x}'y]$, (i) minimizes $E[(y - \mathbf{x}\gamma)^2]$ and (ii) maximizes the correlation between y and $\mathbf{x}\gamma$.

³We did not discuss this in the class.

Proof. Part (i) is Property LP.6 in W. For part (ii), we assume that y has mean 0 and $\mathbf{x}$ has mean $\mathbf{0}$, because a correlation coefficient does not change when the means of the variables are altered. By part (i), for any constant c and any constant vector γ ,

$$E[(y - \mathbf{x}\beta)^2] \leq E[(y - c\mathbf{x}\gamma)^2]. \quad (\text{A.1})$$

Expanding (A.1), we have

$$E[y^2] + E[(\mathbf{x}\beta)^2] - 2E[y\mathbf{x}\beta] \leq E[y^2] + c^2 E[(\mathbf{x}\gamma)^2] - 2cE[y\mathbf{x}\gamma]. \quad (\text{A.2})$$

It follows from this that we can write

$$2 \frac{E[y\mathbf{x}\beta]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\beta)^2]}} - \frac{E[(\mathbf{x}\beta)^2]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\beta)^2]}} \geq 2c \frac{E[y\mathbf{x}\gamma]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\gamma)^2]}} - c^2 \frac{E[(\mathbf{x}\gamma)^2]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\gamma)^2]}}. \quad (\text{A.3})$$

Choose

$$c^2 = \frac{E[(\mathbf{x}\beta)^2]}{E[(\mathbf{x}\gamma)^2]}.$$

Then (A.3) becomes

$$\frac{E[y\mathbf{x}\beta]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\beta)^2]}} \geq \frac{E[y\mathbf{x}\gamma]}{\sqrt{E[y^2]}\sqrt{E[(\mathbf{x}\gamma)^2]}}.$$

□